IE479 - Distribution Logistics

Facility Planning for UMKE Field Health Facilities

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1 Introduction

This report aims to address the strategic facility location problem for UMKE's Field Health Facilities in the aftermath of a large-scale earthquake in Turkey. Using population and distance data for affected districts, this report analyzes three mathematical optimization models to explore the given case questions: the Set Covering Model, the Max Covering Model, and the P-median Model. Each model is chosen to respond to a different planning objective, such as ensuring full coverage, maximizing the population served under limited resources, and minimizing the total travel distance.

2 Methodology and Results

2.1 Q1 - Set Covering Model

For Question 1, each UMKE Field Health Facility must be established near a major road or a district to ensure accessibility for ambulances and rescue vehicles. This is modeled as a Set Covering Model, where the objective is to determine the minimum number of facilities required so that each district is covered by at least one facility within the service distance.

In this model, each candidate site (location) $j \in J$ has a coverage region R. A district $i \in I$ is considered covered if it is within the distance R of at least one selected facility.

Parameters

d_{ij} : Distance (km) between district i and location j

R : Service radius (in km), R= {2, 3, 4, 5, 6}

$$a_{ij} : \begin{cases} 1, & \text{if } d_{ij} \leq R, \\ 0, & \text{otherwise.} \end{cases}$$

Decision Variables

$$x_j = \begin{cases} 1, & \text{if a facility is located in location } j, \\ 0, & \text{otherwise.} \end{cases}$$

Model Formulation

$$\begin{aligned} \text{Minimize} \quad & \sum_{i \in I} x_j && \text{(Objective Function)} \\ \text{subject to} \quad & \sum_{j \in J} a_{ij}\, x_j \geq 1, && \forall i \in I \quad (1) \\ & x_j \in \{0, 1\}, && \forall j \in J \quad (2) \end{aligned}$$

Constraints

1. The first constraint is the coverage constraint. It states that every district i must be covered by at least one facility j located within radius R.
2. The second constraint is the binary constraint that defines the decision variables as discrete choices. Each facility can either be opened or not.

Results and Analysis As expected, the number of facilities required to achieve full coverage decreases as the service radius increases. When R=2 km, coverage zones are small and isolated, requiring many facilities to ensure accessibility. Doubling the radius to 4 km cuts the required facility count by more than half, reflecting strong economies of scale in service reach.

Table 1: Results of Maximal Covering Model for Different Service Radii

Service Radius (km)	**Selected Facility Locations**	**Covered Population**	**Coverage (%)**
2	1, 6, 7, 8, 19	475,840	92.93%
3	2, 6, 9	512,016	100.0%
4	4, 16	512,016	100.0%
5	7	512,016	100.0%
6	5	512,016	100.0%

2.2 Q2 - Max Covering Model

For Question 2, UMKE must prioritize locations that collectively serve the greatest share of the population within a fixed resource limit of 4 facilities at most. This tradeoff is modeled as the Maximal Coverage Location Problem (MCLP), which aims to identify a subset of facility sites that maximizes the total population covered under a given radius for this instance. Unlike the Set Covering Problem that was used in Case 1, MCLP introduces a fixed upper bound on the number of facilities as a budget constraint rather than just minimizing the cost.

In this model, each candidate site (location) $j \in J$ has an associated coverage region defined by the radius of medical service R. A district $i \in I$ is considered covered if it is within the distance R of at least one selected facility.

Parameters

h_i : Population in district i

d_{ij} : Distance (km) between location j and district i

R : Service radius (in km), R= {2, 3, 4, 5, 6}

$$a_{ij} : \begin{cases} 1, & \text{if } d_{ij} \leq R, \\ 0, & \text{otherwise.} \end{cases}$$

p : Maximum number of facilities allowed, here it is set as $p = 4$

Decision Variables

$$y_j = \begin{cases} 1, & \text{if a facility is opened in location } j, \\ 0, & \text{otherwise.} \end{cases}$$

$$x_i = \begin{cases} 1, & \text{if district } i \text{ is covered by at least one open facility,} \\ 0, & \text{otherwise.} \end{cases}$$

Model Formulation

$$
\begin{aligned}
\text{Minimize} \quad & \sum_{i \in I} x_i \, h_i && \text{(Objective Function)} \\
\text{subject to} \quad & \sum_{j \in J} y_j \leq p && (1) \\
& \sum_{j \in J} a_{ij} \, y_j \geq x_i, && \forall i \in I \quad (2) \\
& x_i \in \{0, 1\}, && \forall i \in I \quad (3) \\
& y_j \in \{0, 1\}, && \forall j \in J \quad (4)
\end{aligned}
$$

Constraints

1. The first constraint introduces an upper bound for the number of facilities that can be established to a maximum of four. It reflects the budget constraint that was established by UMKE. In some cases, the optimal solution may use fewer than four facilities if full coverage is already achieved.

2. The second constraint ensures that a district can only be considered covered if there is at least one facility within its coverage radius R.The parameter a_{ij} equals 1 when the distance between district i and location j is less than or equal to R; otherwise, it is 0. This relationship links coverage decisions x_i to facility placement decisions y_j.

3. The third and fourth constraints are binary constraints that define the decision variables as discrete choices. Each facility can either be opened or not, and each district can either be covered or not.

Results and Analysis

The model was applied for service radii $R = 2, 3, 4, 5, 6$ km with a maximum of four facilities. The results for each radius scenario are summarized below. As the radius increases, each facility can serve a wider region, leading to higher total coverage. The full coverage is reached after increasing the service radius to 3 km. Different from other questions, Road 3 is chosen as one of the facility locations, which was an expected result, as the service radius of 2 km is inherently shorter than most distances between districts and major roads.

Table 2: Results of Maximal Covering Model for Different Service Radii

Service Radius (km)	**Selected Facility Locations**	**Covered Population**	**Coverage (%)**
2	6, 7, 8, Road 3	475,840	92.93%
3	1, 2, 6	512,016	100.0%
4	9, 10, 16, 18	512,016	100.0%
5	7	512,016	100.0%
6	5	512,016	100.0%

2.3 Q3 - Absolute P-Center Model

In Question 3, UMKE aims to minimize the overall travel distance between locations and their assigned health facilities while ensuring that each district is within the specified service radius. However, the model should select 4 facilities to do so.

In this model, as in the previous two models, facilities could also be placed on the nearest roads to ensure faster patient transport and improved response times. Each candidate site (location) $j \in J$ has an associated coverage region defined by the radius of medical service R. A district $i \in I$ is considered covered if it is within the distance R of at least one selected facility.

Parameters

h_i : Population of district i

d_{ij} : Distance (km) between location j and district i

R : Service radius (in km), R= {2, 3, 4, 5, 6}

$$a_{ij} : \begin{cases} 1, & \text{if } d_{ij} \leq R, \\ 0, & \text{otherwise.} \end{cases}$$

p : Maximum number of facilities allowed, here it is set as $p = 4$

Decision Variables

$$x_{ij} = \begin{cases} 1, & \text{if district } i \text{ is covered by open facility } j, \\ 0, & \text{otherwise.} \end{cases}$$

$$y_j = \begin{cases} 1, & \text{if a facility is open at location } j, \\ 0, & \text{otherwise.} \end{cases}$$

Model Formulation

$$
\begin{aligned}
\min \quad & \sum_{i \in I} \sum_{j \in J} h_i \, d_{ij} \, x_{ij} && \text{(Objective Function)} \\
\text{subject to} \quad & \sum_{j \in J} x_{ij} = 1, && \forall i \in I \quad (1) \\
& x_{ij} \leq a_{ij}, && \forall i \in I, j \in J \quad (2) \\
& x_{ij} \leq y_j, && \forall i \in I, j \in J \quad (3) \\
& \sum_{j \in J} y_j = p, && (4) \\
& x_{ij}, y_j \in \{0, 1\}, && \forall i \in I, j \in J \quad (5)
\end{aligned}
$$

Constraints

1. The first constraint ensures that each district i is assigned to exactly one facility j. This guarantees that every district receives service from one UMKE Facility.
2. The second constraint ensures that a district can only be considered covered if there is at least one facility within its coverage radius R. This constraint prevents any district from being assigned to a facility that is too far away.
3. The third constraint prevents districts from being assigned to an unopened facility.
4. The fourth constraint limits the total number of facilities that can be opened to the dedicated upper limit.
5. The fifth constraint defines the decision variables as binary. Each facility can either be opened or not, and each district can either be covered or not.

Results and Analysis

The model was applied for service radii $R = 3, 4, 5, 6$ km with a number of four facilities. The results for each radius scenario are summarized below. The radius increase does not affect the selected facility locations. All of these solutions give 22.21 km for the total travel distance between a facility and a district for all radii tested. The total population-weighted distance for each radius is 618,545 person-km. This also does not change for each radius because the same solution applies to each scenario.

Table 3: Absolute P-Center Model for Different Service Radii

Service Radius (km)	**Selected Facilities**	**Max Distance (km)**	**Total Distance (km)**
3	1, 2, 12, 15	2.31	22.21
4	1, 2, 12, 15	2.31	22.21
5	1, 2, 12, 15	2.31	22.21
6	1, 2, 12, 15	2.31	22.21